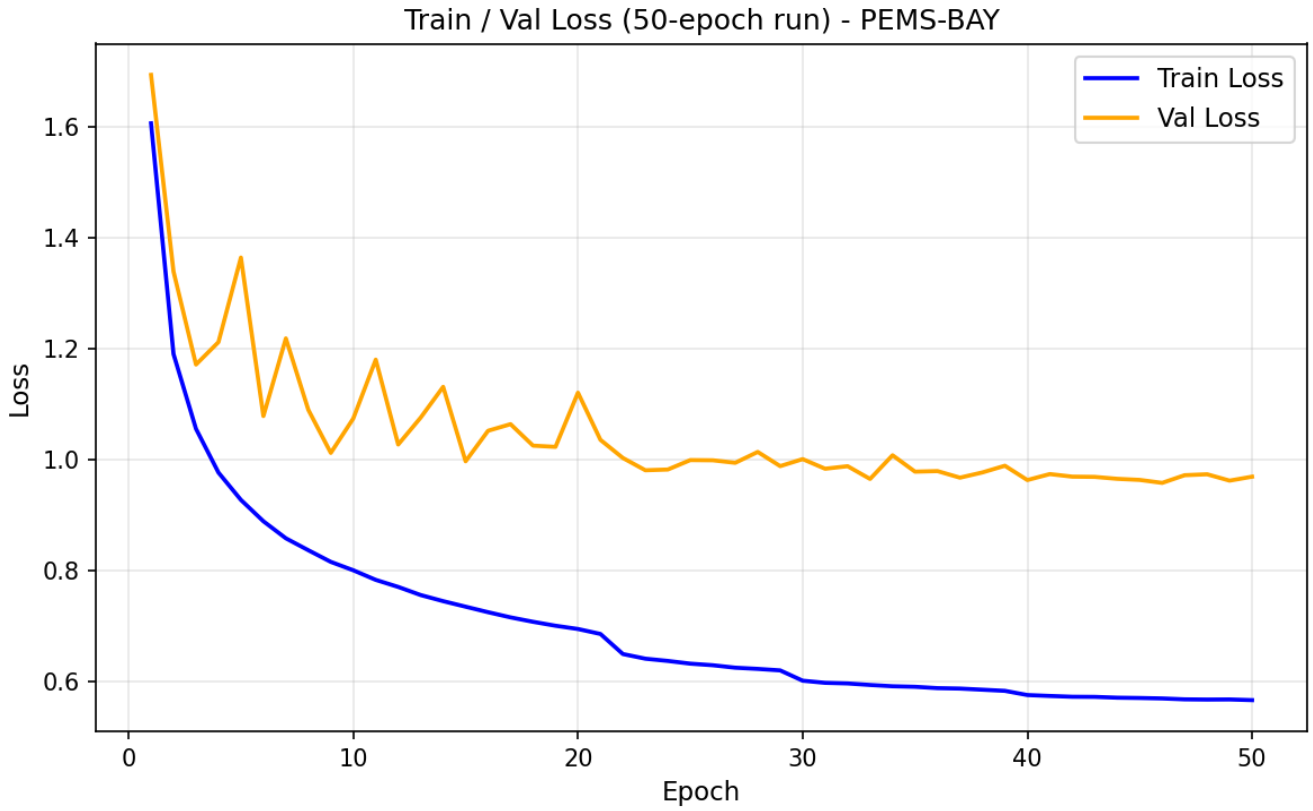
## Model and training configuration:

| Component | Specification |
|---|---|
| **Model Architecture** | |
| Dataset | PEMS-BAY |
| Number of Sensors | 325 |
| Input Window | 12 time steps (60 minutes, 5-min intervals) |
| Prediction Horizon | 12 time steps (60 minutes ahead) |
| Spatio-Temporal Blocks | 4 stacked graph/spatio-temporal layers |
| Hidden Feature Dimension | 64 |
| Diffusion Steps | 2-hop propagation |
| Graph Attention Heads | 4 |
| Temporal Convolution | Kernel size 2; dilations {1, 2, 4} |
| BiLSTM Hidden Size | 64 |
| Dropout Rate | 0.15 |
| Total Parameters | ≈270K (270,340) |
| **Training Setup** | |
| Optimizer | AdamW (learning rate = $10^{-3}$, weight decay = $10^{-5}$) |
| Scheduler | ReduceLROnPlateau (factor = 0.5, patience = 5) |
| Epochs / Batch Size | 50 / 8 |
| Loss Function | Uncertainty-aware loss: MSE + NLL |
| Gradient Clipping | max_norm = 1.0 |
| Early Stopping | Patience = 15 epochs |
| **Uncertainty Estimation** | |
| MC Dropout | Enabled at inference |
| MC Samples | 10 stochastic forward passes |
| Aleatoric Uncertainty | Learned through the variance prediction head |
| Epistemic Uncertainty | Estimated from variance across MC forward passes |

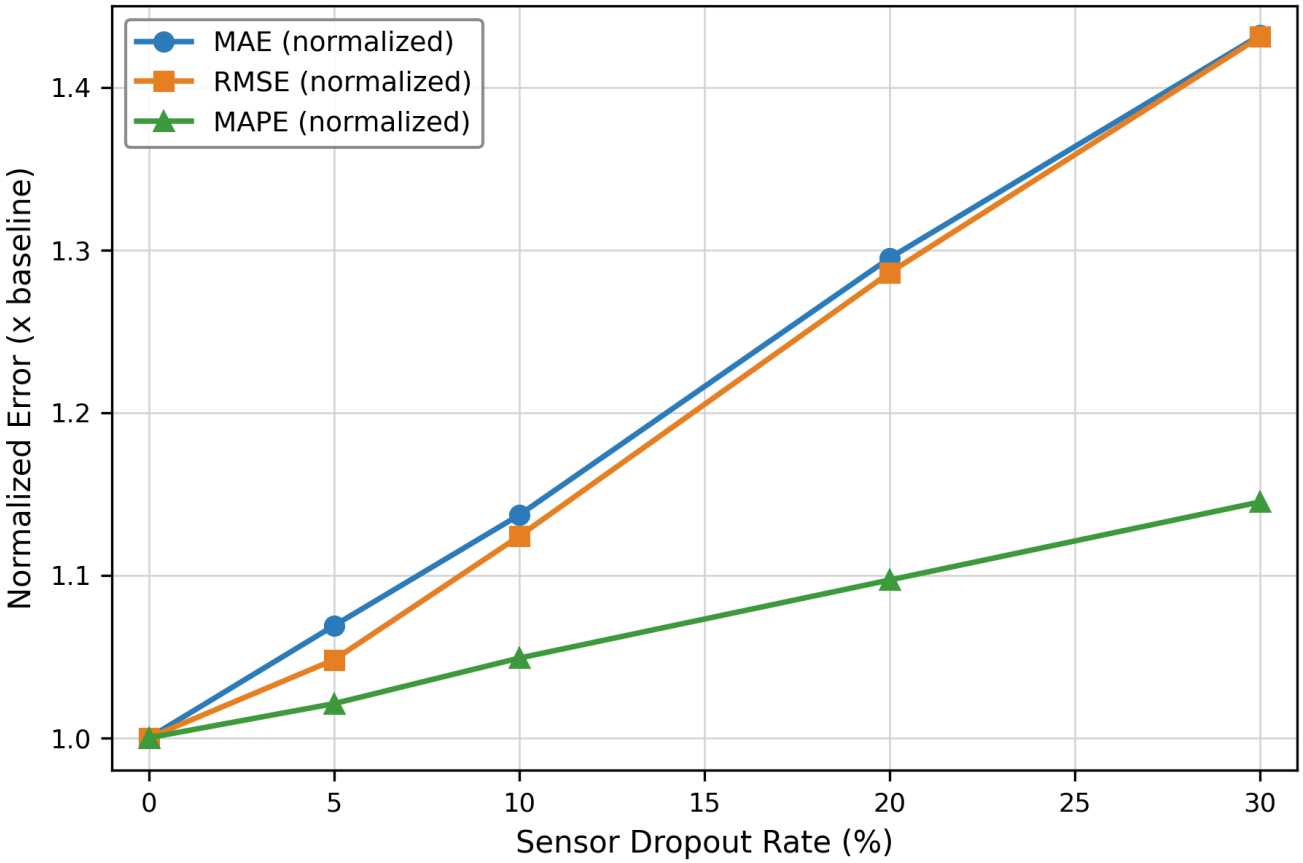
## Performance analysis on PEMS-BAY test set:

| Metric | Value |
|---|---|
| MAE | 0.4392 |
| RMSE | 1.0327 |
| R2 Score | 0.8385 |
| Pearson Correlation | 0.9158 |
| Mean Aleatoric Uncertainty | 0.5508 |
| Mean Epistemic Uncertainty | 0.2560 |
| Mean Total Uncertainty | 0.8068 |

Training and validation loss curves over 50epochs, demonstrating smooth convergence and stable gen-eralization behavior on the PEMS-BAY dataset.



Robustness of UA-GNN under sensor dropout on the PEMS-BAY dataset



Correlation matrix of a sensor subset, motivating graph-based spatial modeling.

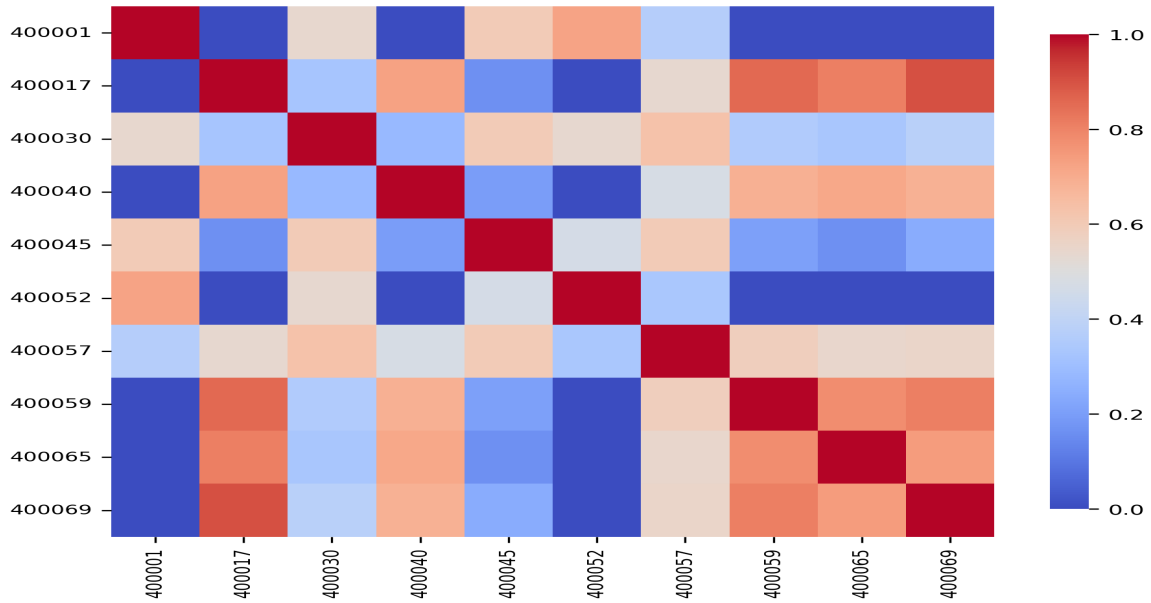



Figure 6: Correlation matrix of a sensor subset, motivating graph-based spatial modeling.

Representative 24-hour traffic speed profile with predictive uncertainty band. Wider intervals correspond to peak-hour variability, while narrower bands indicate stable traffic conditions.

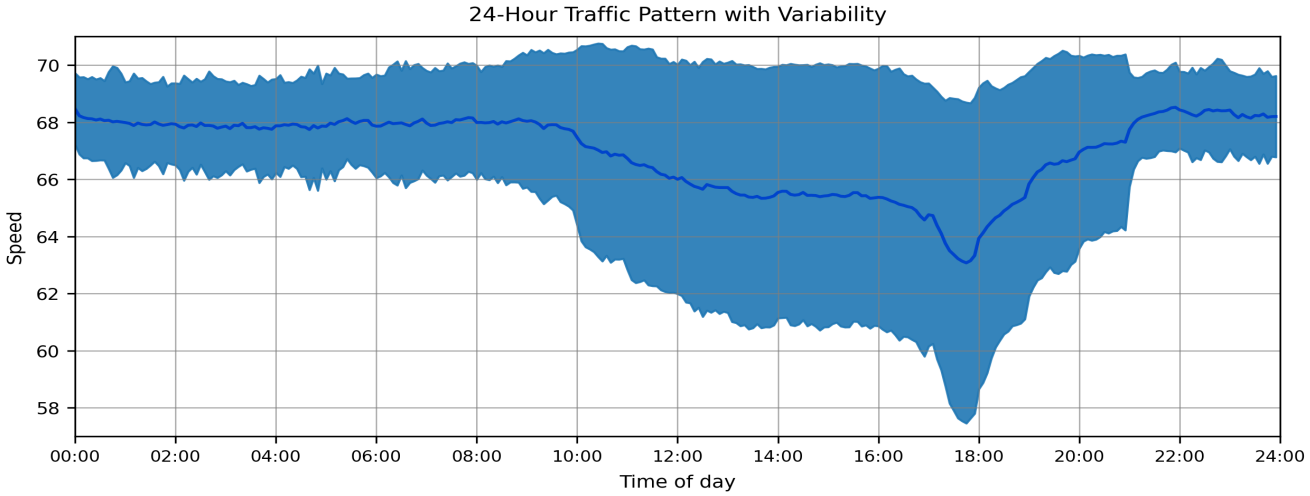



Figure 5: Representative 24-hour traffic speed profile with predictive uncertainty band. Wider intervals correspond to peak-hour variability, while narrower bands indicate stable traffic conditions.

## Ablation Study: Component contribution over-all Performance on PEMS-BAY Dataset:

| Model Variant | MAE | RMSE | MAE Degradation (%) |
|---|---|---|---|
| Full Model (all components) | 0.4392 | 1.0327 | Baseline |
| w/o Diffusion Conv (GCN only) | 23.6171 | 35.7619 | +5277.30% |
| w/o BiLSTM (Temporal Conv only) | 1.5148 | 2.3970 | +244.90% |
| w/o Graph Attention | 1.1911 | 1.8554 | +171.20% |
| w/o MC Dropout (Deterministic) | 0.4392 | 1.0327 | +0.00% |

## Computational Efficiency Comparison on PEMS-BAY:

| Model | Inference Time (ms) | Memory (MB) | Parameters (K) | Notes / Source |
|---|---|---|---|---|
| **Proposed Model** | 59.3 ± 0.7 | 358 (train buffer) / 99.8 (batch=32) / 95.4 ± 1.3 (batch=8 eval) | 270.3 (core) / 303.8 (w/ embeddings) | Empirically profiled on GPU (exact weights: 270,340) |
| DCRNN | 174.0 ± 3.8 | 1,417 | 1,200.0 | 325 graph node scaling from published baseline |
| Graph WaveNet | 96.2 ± 2.4 | 803 | 700.0 | 325 graph node scaling from published baseline |
| Bayesian LSTM | 125.1 ± 3.1 | 1,057 | 900.0 | 325 node scaling from published baseline |

## UA-GNN FINAL EVALUATION SUMMARY (PEMS-BAY DATASET)

**1. OVERALL METRICS (Standardized / Normalized Latent Space):**

- MAE: 0.439200
- RMSE: 1.032700
- $R^2$ Score: 0.838500
- Pearson Correlation: 0.915800
- Mean Aleatoric Uncertainty: 0.550800
- Mean Epistemic Uncertainty: 0.256000
- Mean Total Uncertainty: 0.806800

**2. PHYSICAL SPEED METRICS (Miles Per Hour, mph):**

- Overall MAE: 1.6700 mph
- Overall RMSE: 3.7200 mph
- Overall MAPE: 3.85 %

**3. HORIZON-SPECIFIC METRICS (Physical mph):**

- 15 min (step 3): MAE = 1.3000 mph, RMSE = 2.7400 mph
- 30 min (step 6): MAE = 1.6400 mph, RMSE = 3.7000 mph
- 45 min (step 9): MAE = 1.8800 mph, RMSE = 4.2100 mph
- 60 min (step 12): MAE = 2.0700 mph, RMSE = 4.7400 mph

**4. CALIBRATION:**

- Expected Calibration Error (ECE): 0.0869